OPENSOURCESDRLAB DOCUMENTATION

FMC-ADRV9009 Schematic

Product	FMC-ADRV9009
Document	OSSDRLAB-FMC-ADRV9009-SCH
Version	1.0
Date	2026-11-12

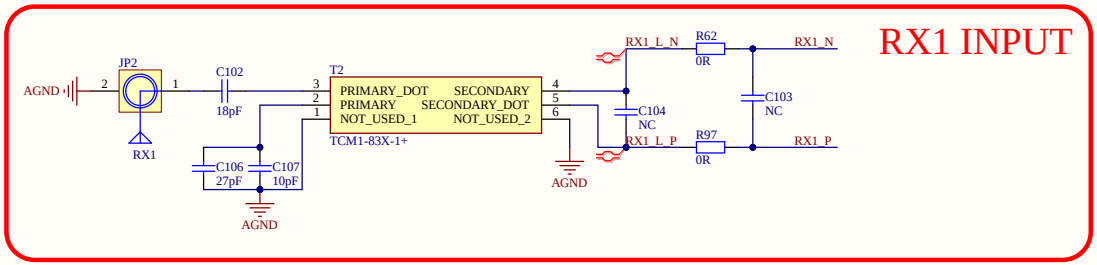
This document contains the circuit schematic of the OpenSourceSDRLab FMC-ADRV9009 board, exported from Altium Designer (A3). The seven schematic sheets follow this cover page.

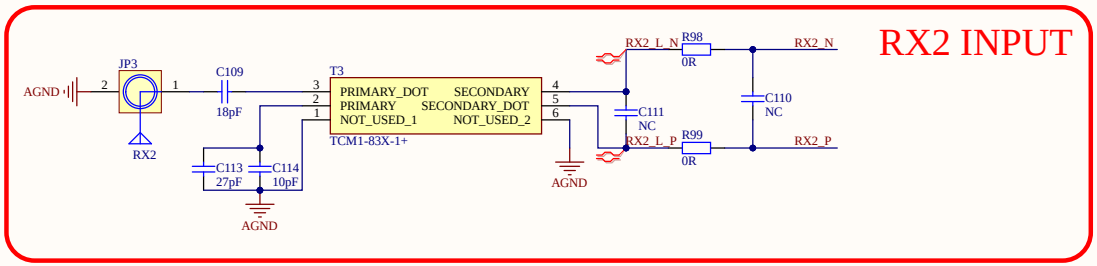
Sheet	Content
1	RF inputs: RX1, RX2, ORX1, ORX2 baluns
2	RF outputs TX1 and TX2, external LO input/output
3	Clocking: AD9528 clock generator, reference clock input, VCXO, status LEDs, SPI level conversion
4	ADRV9009 GPIO interface and JESD204B SERDES lanes
5	ADRV9009 power pins and decoupling
6	Power supply: 12 V input, ADP5054 regulator, filtering, rail enables, clock supply
7	FMC HPC connector

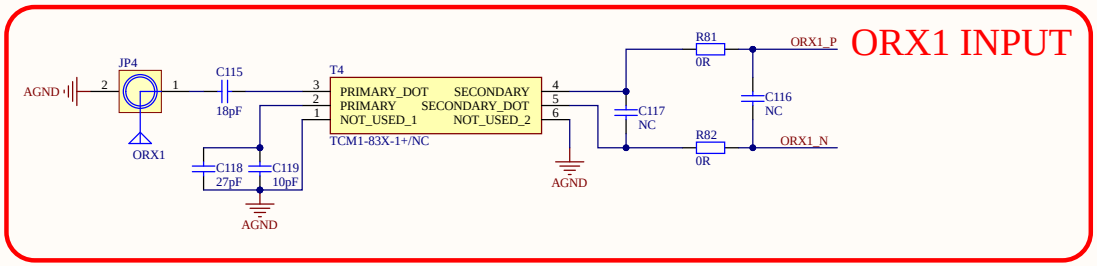
Product page: https://opensourcesdrlab.com/products/opensourcesdrlab-new-fmc-adrv9009-board-high-performance-board

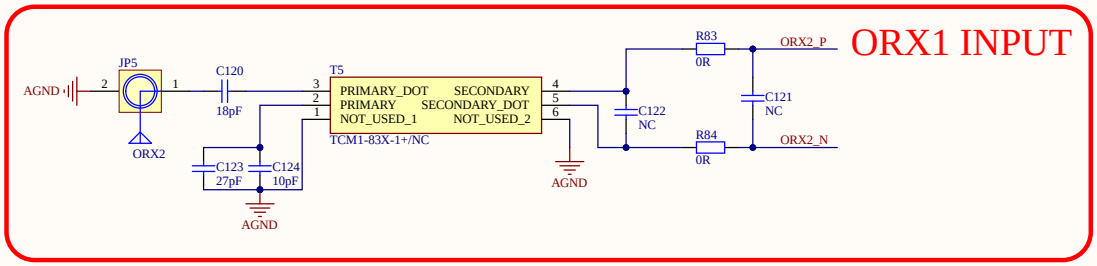
Questions about the hardware: support@opensourcesdrlab.com

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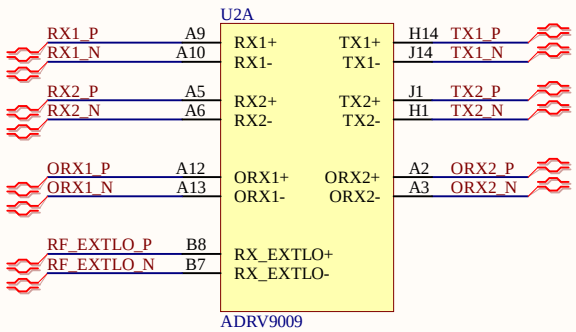


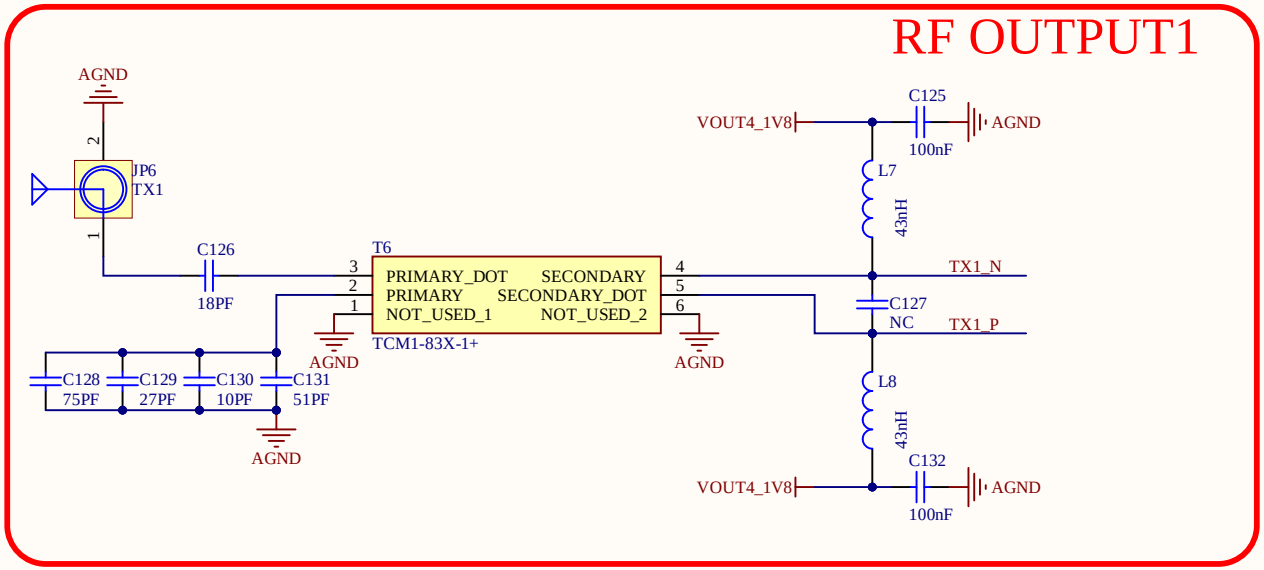


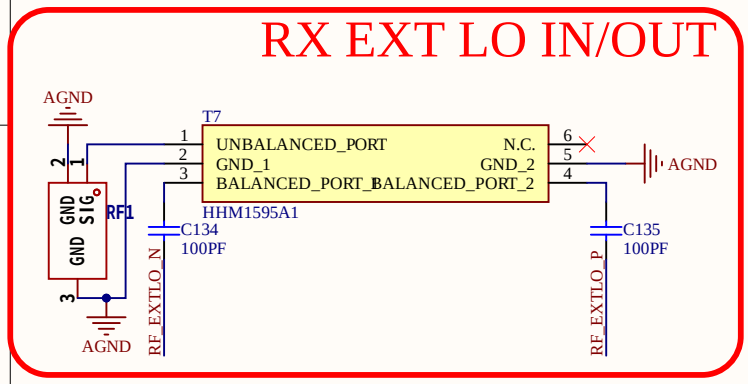


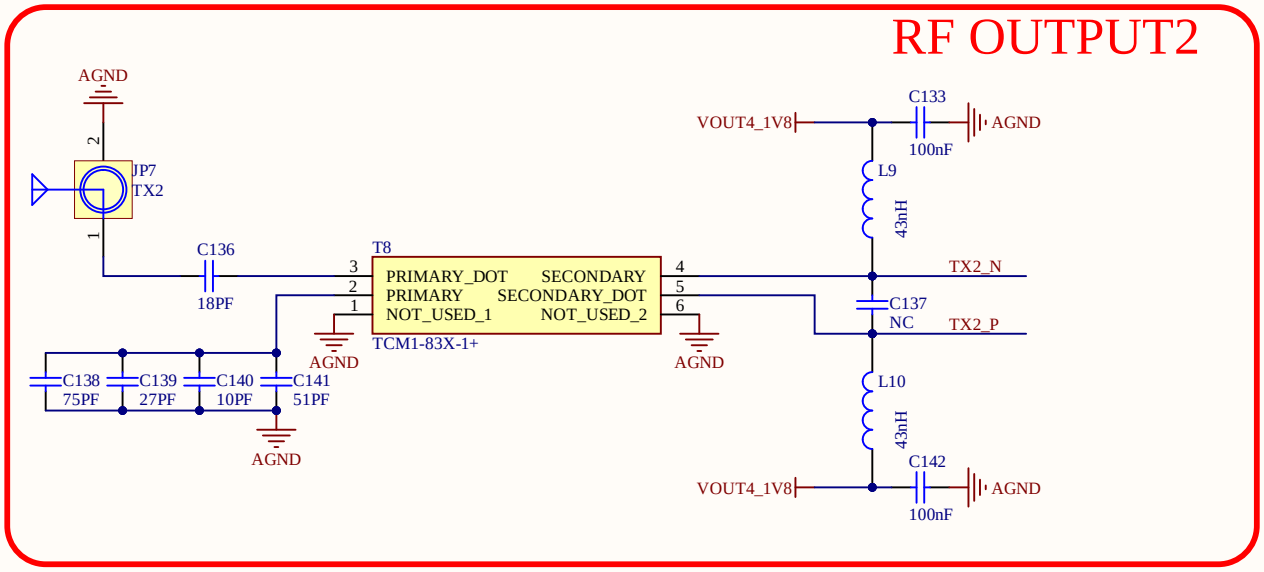


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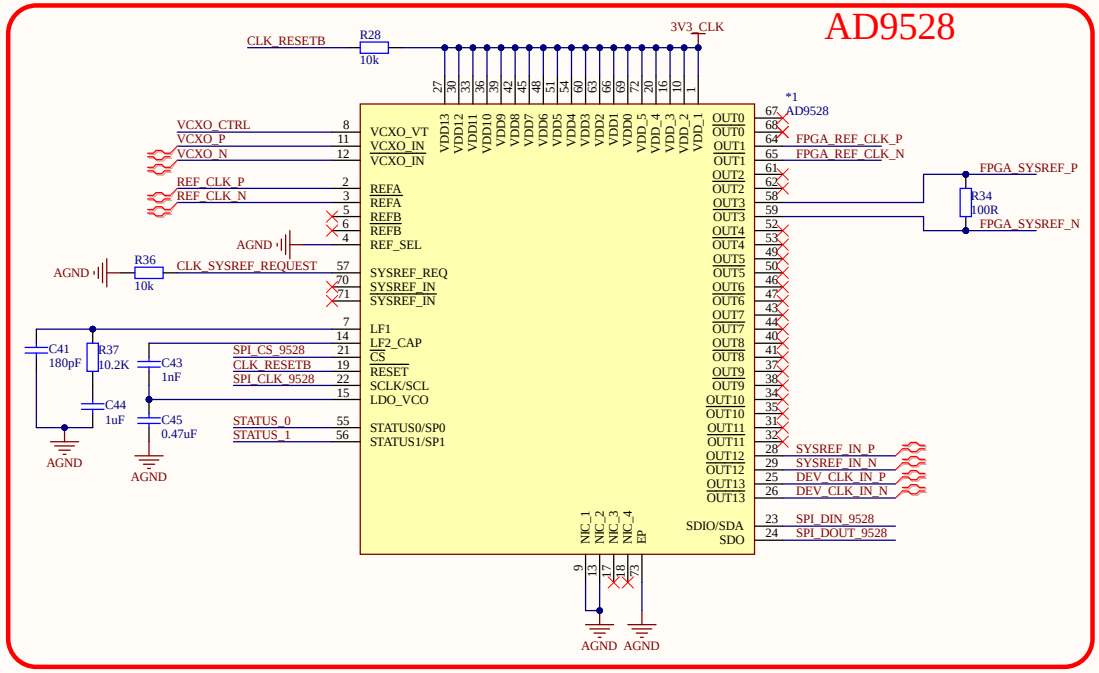


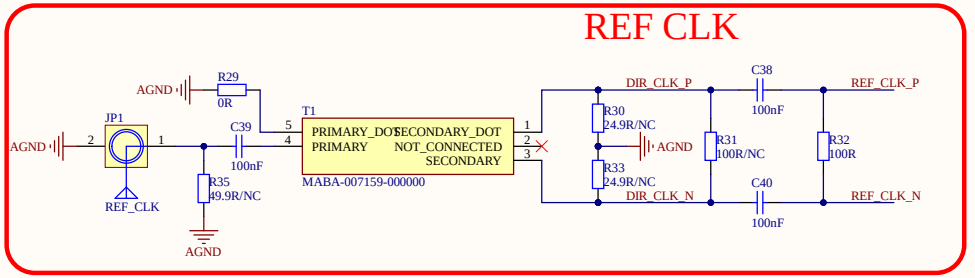


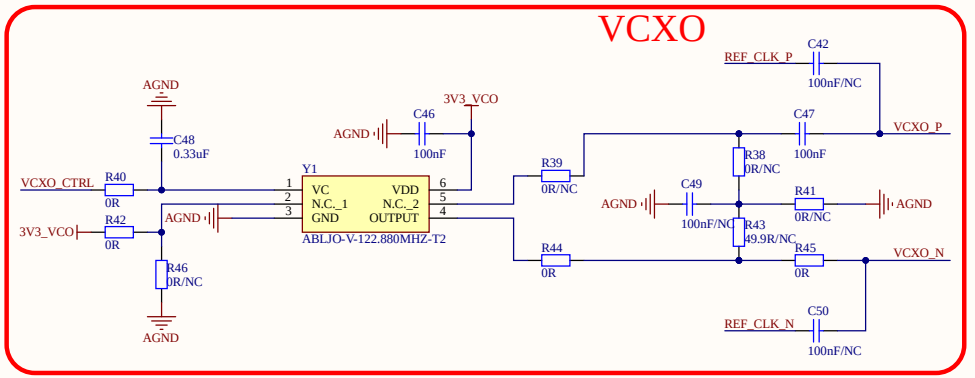


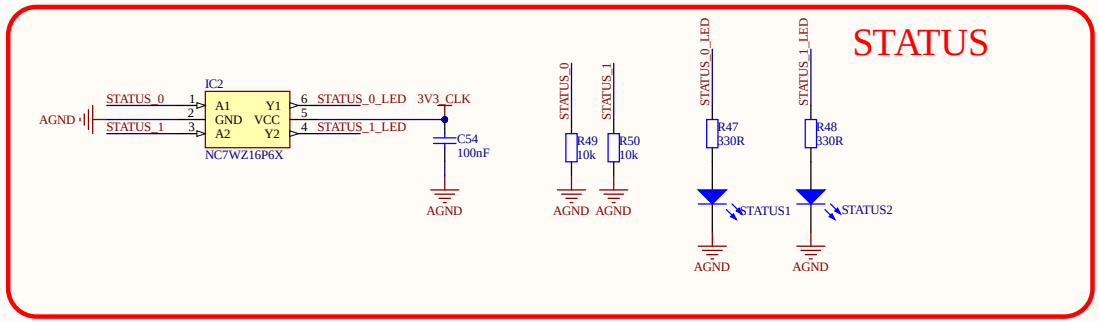


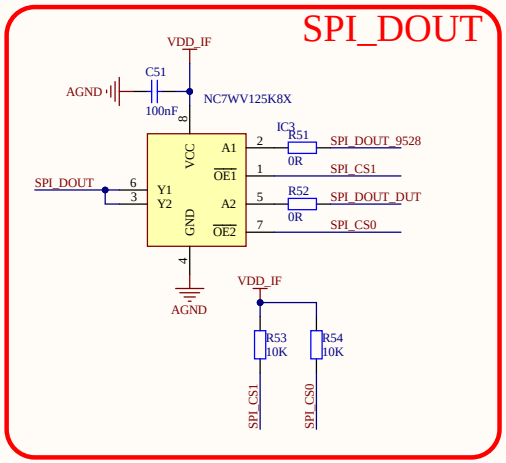
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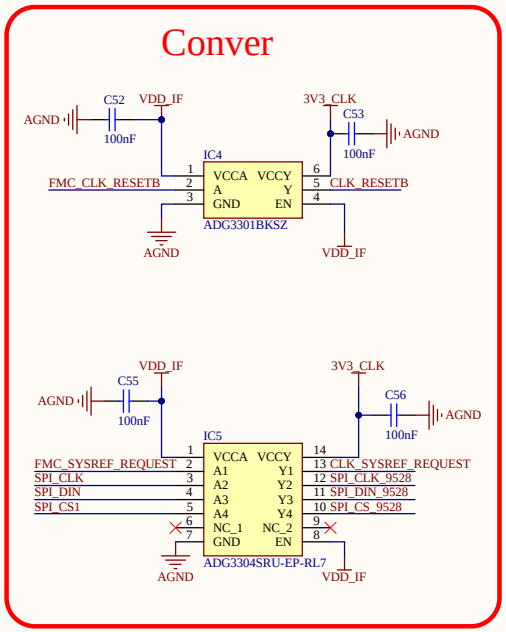


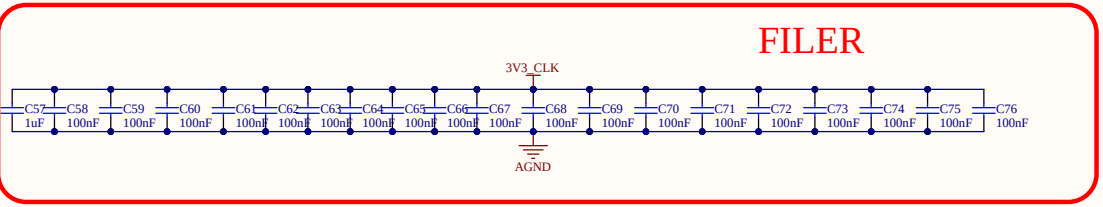




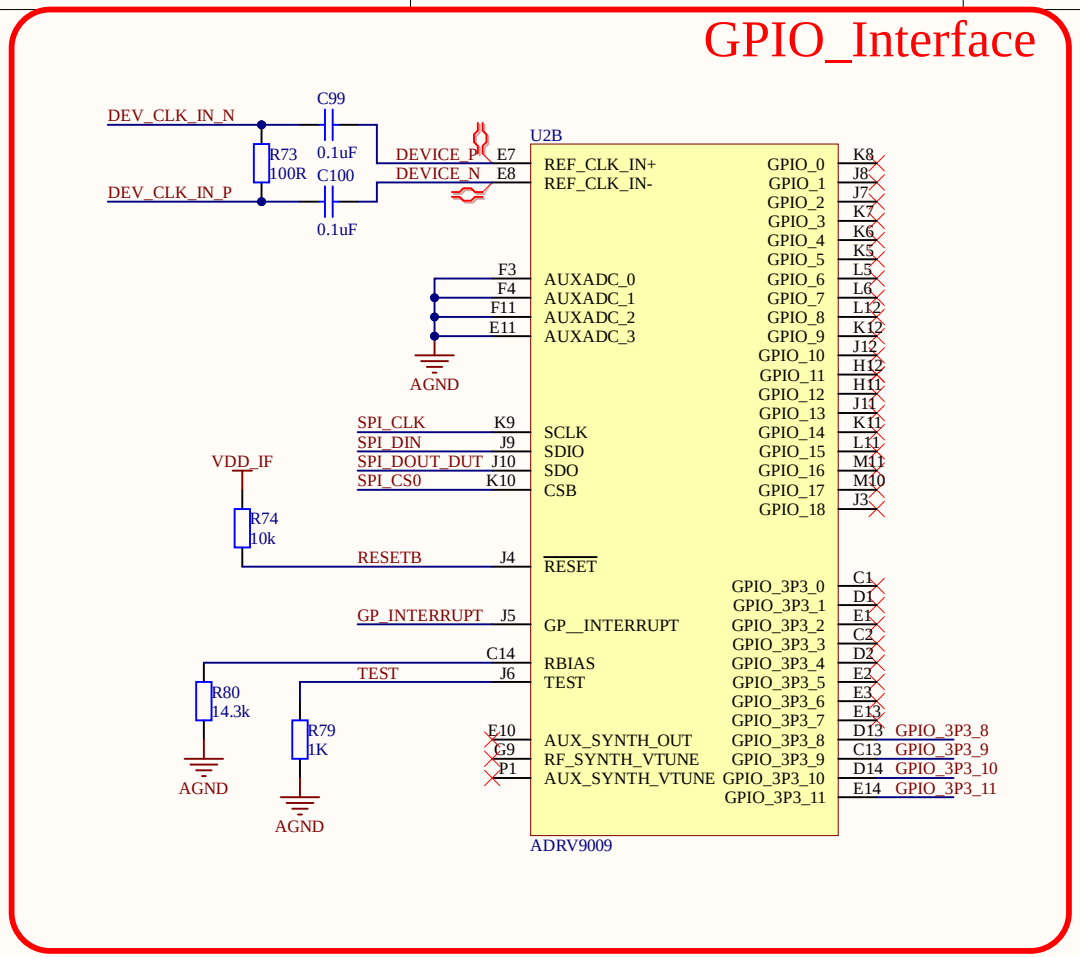


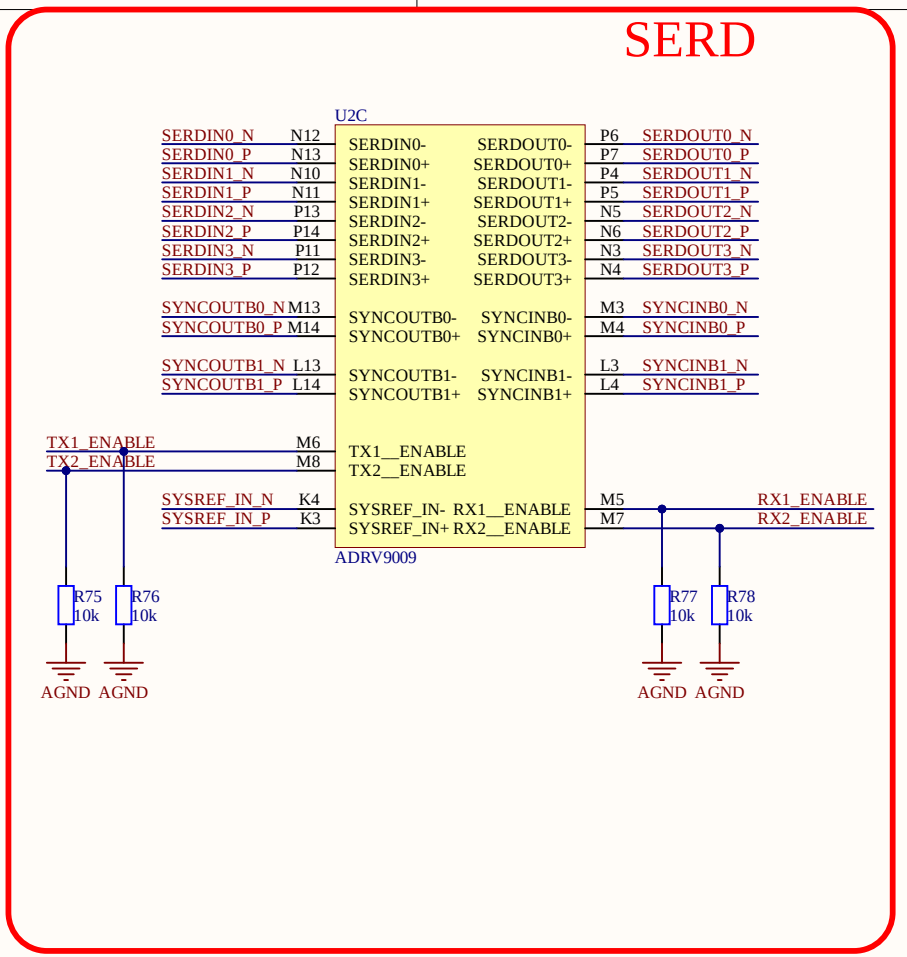




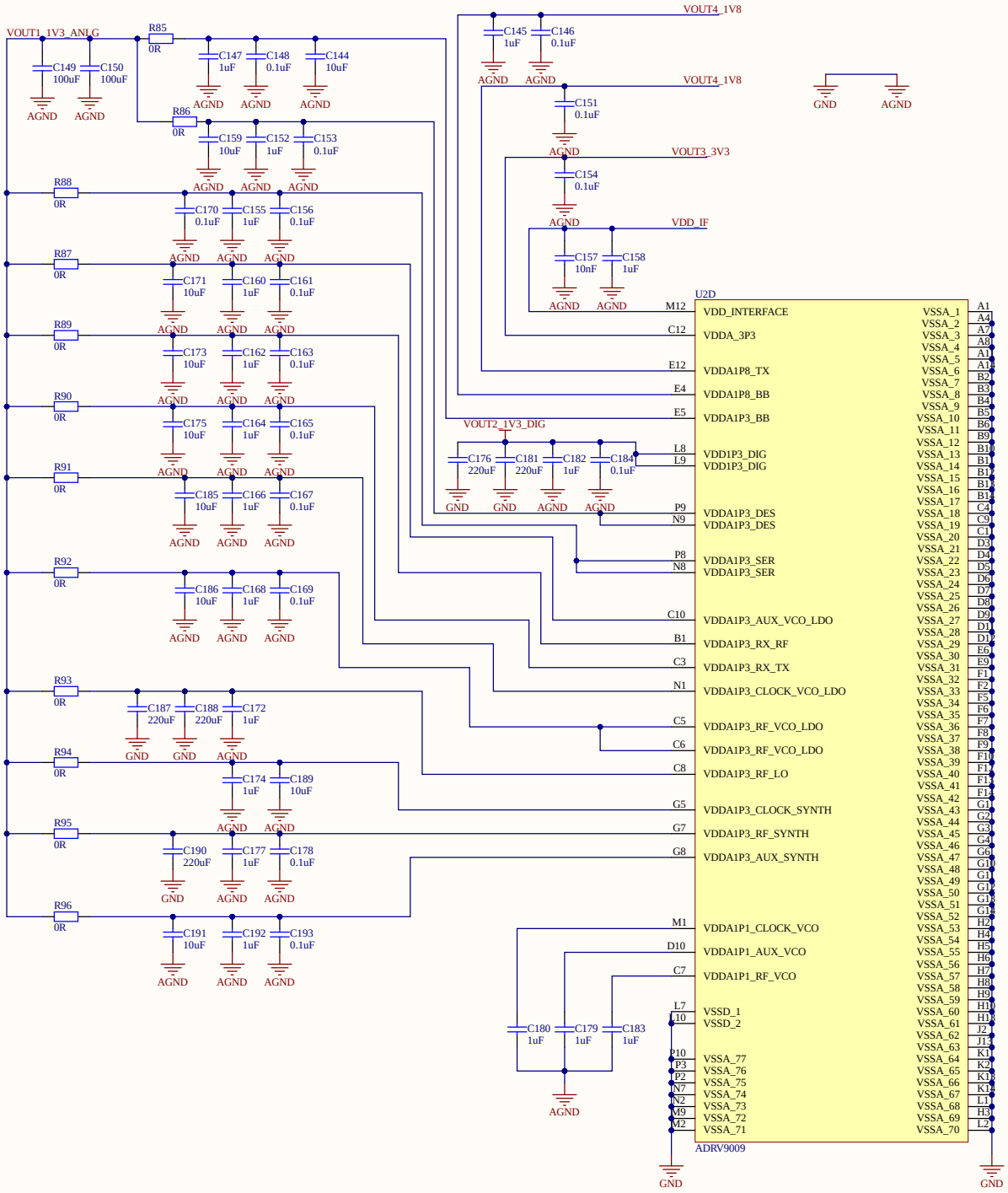


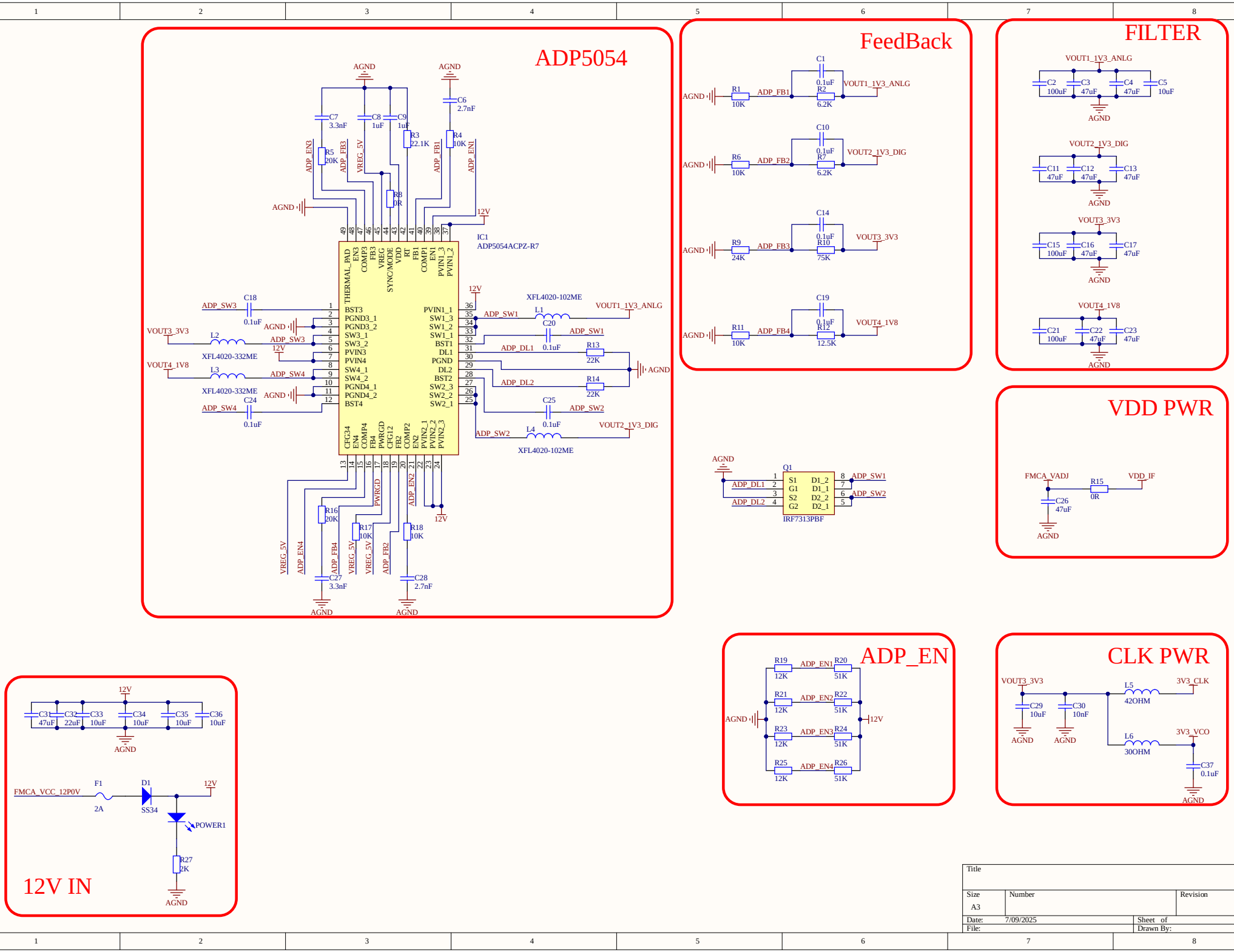
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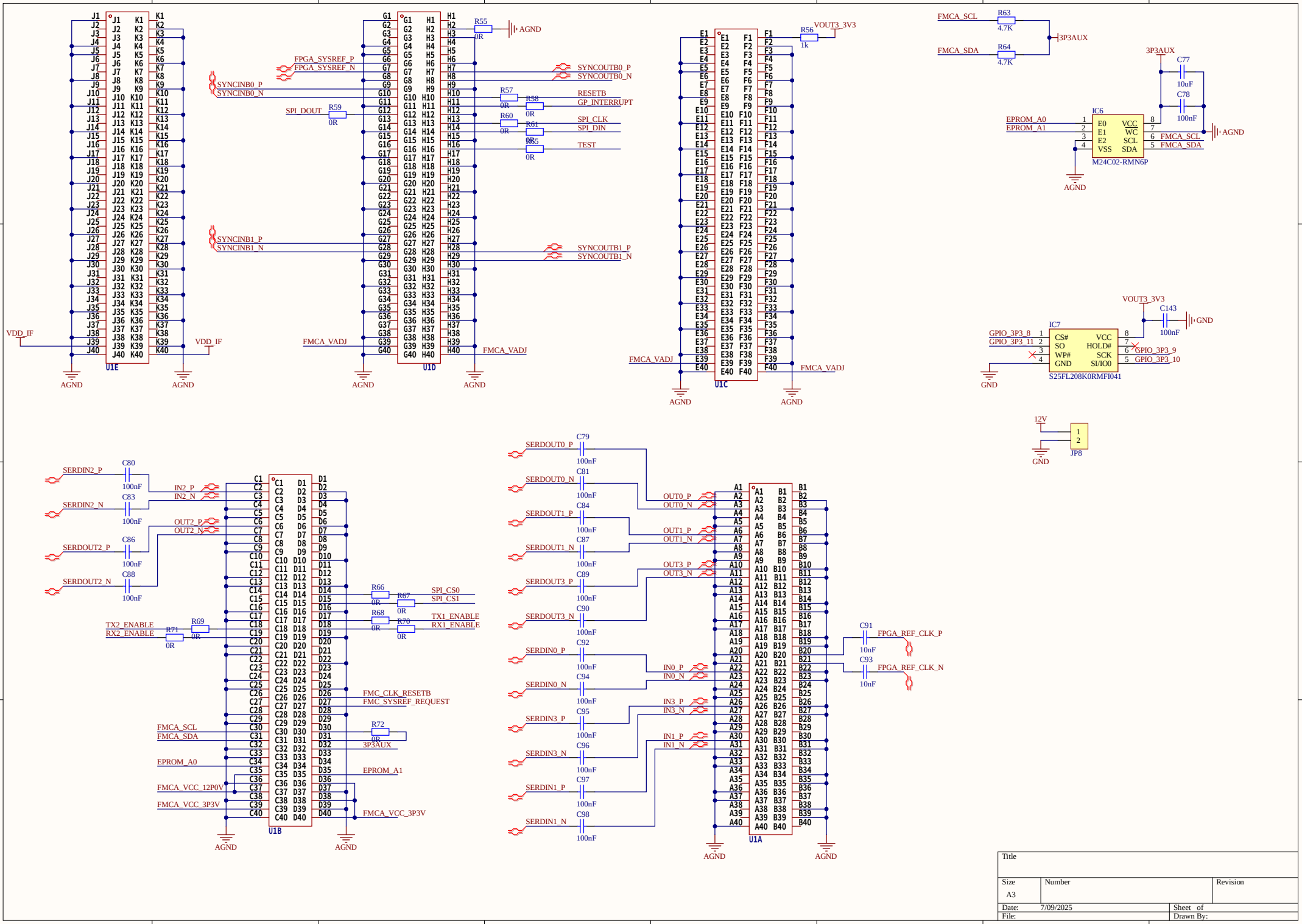




Title		
Size: A4	Number	Revision
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VOUT1_1V3_ANLG
R85
0R
C149 100uF
C150 100uF
C147 1uF
C148 0.1uF
C144 10uF
R86
0R
C159 10uF
C152 1uF
C153 0.1uF
R88
0R
C170 0.1uF
C155 1uF
C156 0.1uF
R87
0R
C171 10uF
C160 1uF
C161 0.1uF
R89
0R
C173 10uF
C162 1uF
C163 0.1uF
R90
0R
C175 10uF
C164 1uF
C165 0.1uF
R91
0R
C185 10uF
C166 1uF
C167 0.1uF
R92
0R
C186 10uF
C168 1uF
C169 0.1uF
R93
0R
C187 220uF
C188 220uF
C172 1uF
R94
0R
C174 1uF
C189 10uF
R95
0R
C190 220uF
C177 1uF
C178 0.1uF
R96
0R
C191 10uF
C192 1uF
C193 0.1uF
AGND
GND
VOUT4_1V8
C145 1uF
C146 0.1uF
C151 0.1uF
VOUT3_3V3
C154 0.1uF
VDD_IF
C157 10nF
C158 1uF
VOUT2_1V3_DIG
C176 220uF
C181 220uF
C182 1uF
C184 0.1uF
C180 1uF
C179 1uF
C183 1uF
U2D
ADRV9009
M12 VDD_INTERFACE
C12 VDDA_3P3
E12 VDDA1P8_TX
E4 VDDA1P8_BB
E5 VDDA1P3_BB
L8 VDD1P3_DIG
L9 VDD1P3_DIG
P9 VDDA1P3_DES
N9 VDDA1P3_DES
P8 VDDA1P3_SER
N8 VDDA1P3_SER
C10 VDDA1P3_AUX_VCO_LDO
B1 VDDA1P3_RX_RF
C3 VDDA1P3_RX_TX
N1 VDDA1P3_CLOCK_VCO_LDO
C5 VDDA1P3_RF_VCO_LDO
C6 VDDA1P3_RF_VCO_LDO
C8 VDDA1P3_RF_LO
G5 VDDA1P3_CLOCK_SYNTH
G7 VDDA1P3_RF_SYNTH
G8 VDDA1P3_AUX_SYNTH
M1 VDDA1P1_CLOCK_VCO
D10 VDDA1P1_AUX_VCO
C7 VDDA1P1_RF_VCO
L7 VSSD_1
L10 VSSD_2
P10 VSSA_77
P3 VSSA_76
P2 VSSA_75
N7 VSSA_74
N2 VSSA_73
M9 VSSA_72
M2 VSSA_71
VSSA_1 A1
VSSA_2 A4
VSSA_3 A7
VSSA_4 A8
VSSA_5 A11
VSSA_6 A12
VSSA_7 B2
VSSA_8 B3
VSSA_9 B4
VSSA_10 B5
VSSA_11 B6
VSSA_12 B9
VSSA_13 B11
VSSA_14 B12
VSSA_15 B13
VSSA_16 B14
VSSA_17 C4
VSSA_18 C9
VSSA_19 C11
VSSA_20 D3
VSSA_21 D4
VSSA_22 D5
VSSA_23 D6
VSSA_24 D7
VSSA_25 D8
VSSA_26 D9
VSSA_27 D11
VSSA_28 D12
VSSA_29 E6
VSSA_30 E9
VSSA_31 F1
VSSA_32 F2
VSSA_33 F5
VSSA_34 F6
VSSA_35 F7
VSSA_36 F8
VSSA_37 F9
VSSA_38 F10
VSSA_39 F11
VSSA_40 F12
VSSA_41 F13
VSSA_42 G1
VSSA_43 G2
VSSA_44 G3
VSSA_45 G4
VSSA_46 G6
VSSA_47 G10
VSSA_48 G11
VSSA_49 G12
VSSA_50 G13
VSSA_51 G14
VSSA_52 H2
VSSA_53 H4
VSSA_54 H5
VSSA_55 H6
VSSA_56 H7
VSSA_57 H8
VSSA_58 H9
VSSA_59 H10
VSSA_60 H11
VSSA_61 J2
VSSA_62 J11
VSSA_63 K1
VSSA_64 K2
VSSA_65 K11
VSSA_66 K12
VSSA_67 L1
VSSA_68 L3
VSSA_69 L12
VSSA_70 L2
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ADP5054
IC1
ADP5054ACPZ-R7
THERMAL_PAD
BST3
PGND3_1
PGND3_2
SW3_1
SW3_2
PVIN3
PVIN4
SW4_1
SW4_2
PGND4_1
PGND4_2
BST4
PVIN1_1
SW1_3
SW1_2
SW1_1
BST1
DL1
PGND
DL2
BST2
SW2_3
SW2_2
SW2_1
CFG34
EN4
COMP4
FB4
PWRGD
CFG12
FB2
COMP2
EN2
PVIN2_1
PVIN2_2
PVIN2_3
EN3
COMP3
FB3
VREG
SYNC/MODE
VDD
RT
FB1
COMP1
EN1
PVIN1_3
PVIN1_2
AGND
12V
VREG_5V
ADP_EN2
ADP_FB3
ADP_FB1
ADP_EN1
ADP_SW3
ADP_SW4
ADP_SW1
ADP_SW2
ADP_DL1
ADP_DL2
ADP_FB4
ADP_FB2
ADP_EN4
ADP_EN3
PWRGD
C6 2.7nF
C7 3.3nF
C8 1uF
C9 1uF
R3 22.1K
R4 10K
R5 20K
R8 0R
C18 0.1uF
C24 0.1uF
C20 0.1uF
C25 0.1uF
L1 XFL4020-102ME
L2 XFL4020-332ME
L3 XFL4020-332ME
L4 XFL4020-102ME
R13 22K
R14 22K
R16 20K
R17 10K
R18 10K
C27 3.3nF
C28 2.7nF
VOUT1_1V3_ANLG
VOUT2_1V3_DIG
VOUT3_3V3
VOUT4_1V8
FeedBack
C1 0.1uF
R1 10K
R2 6.2K
C10 0.1uF
R6 10K
R7 6.2K
C14 0.1uF
R9 24K
R10 75K
C19 0.1uF
R11 10K
R12 12.5K
FILTER
C2 100uF
C3 47uF
C4 47uF
C5 10uF
C11 47uF
C12 47uF
C13 47uF
C15 100uF
C16 47uF
C17 47uF
C21 100uF
C22 47uF
C23 47uF
VDD PWR
FMCA_VADJ
VDD_IF
R15 0R
C26 47uF
Q1
IRF7313PBF
S1
G1
S2
G2
D1_2
D1_1
D2_2
D2_1
ADP_EN
R19 12K
R20 51K
R21 12K
R22 51K
R23 12K
R24 51K
R25 12K
R26 51K
CLK PWR
3V3_CLK
3V3_VCO
L5 42OHM
L6 30OHM
C29 10uF
C30 10nF
C37 0.1uF
12V IN
FMCA_VCC_12P0V
F1 2A
D1 SS34
POWER1
R27 2K
C31 47uF
C32 22uF
C33 10uF
C34 10uF
C35 10uF
C36 10uF
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U1E
U1D
U1C
U1B
U1A
VDD_IF
AGND
FPGA_SYSREF_P
FPGA_SYSREF_N
SYNCINB0_P
SYNCINB0_N
SYNCINB1_P
SYNCINB1_N
SPI_DOUT
SYNCOUTB0_P
SYNCOUTB0_N
SYNCOUTB1_P
SYNCOUTB1_N
RESETB
GP_INTERRUPT
SPI_CLK
SPI_DIN
TEST
FMCA_VADJ
VOUT3_3V3
FMCA_SCL
FMCA_SDA
3P3AUX
EPROM_A0
EPROM_A1
IC6
M24C02-RMN6P
IC7
S25FL208K0RMFI041
GPIO_3P3_8
GPIO_3P3_11
GPIO_3P3_9
GPIO_3P3_10
GND
12V
JP8
SERDIN2_P
SERDIN2_N
SERDOUT2_P
SERDOUT2_N
IN2_P
IN2_N
OUT2_P
OUT2_N
TX2_ENABLE
RX2_ENABLE
SPI_CS0
SPI_CS1
TX1_ENABLE
RX1_ENABLE
FMC_CLK_RESETB
FMC_SYSREF_REQUEST
FMCA_VCC_12P0V
FMCA_VCC_3P3V
SERDOUT0_P
SERDOUT0_N
SERDOUT1_P
SERDOUT1_N
SERDOUT3_P
SERDOUT3_N
SERDIN0_P
SERDIN0_N
SERDIN3_P
SERDIN3_N
SERDIN1_P
SERDIN1_N
OUT0_P
OUT0_N
OUT1_P
OUT1_N
OUT3_P
OUT3_N
IN0_P
IN0_N
IN3_P
IN3_N
IN1_P
IN1_N
FPGA_REF_CLK_P
FPGA_REF_CLK_N
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